Software Requirements Specification
NYC Traffic Accident Cluster Discovery System in R

AMS 597 Project Team

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1 Purpose

This project proposes an R-based analytical system that discovers meaningful patterns in New York City traffic accidents using clustering techniques. The system is not designed as a purely descriptive dashboard and it is not primarily a supervised prediction system. Its purpose is to discover latent accident patterns, accident scenario groups, and geographic hotspot structures that are not obvious from simple count tables.

The project is restricted to New York City only in order to keep the scope focused, produce a more original analysis, and make the final report easier to defend academically.

2 Problem Statement

The NYC accident data contains mixed variable types:

- numeric variables such as temperature, visibility, wind speed, and pressure
- categorical variables such as weather condition and day versus night
- binary roadway context variables such as `Crossing`, `Junction`, and `Traffic_Signal`
- temporal information derivable from timestamps such as hour band, weekday, season, and rush-hour status

Because of this structure, the main research problem is:

How can we discover interpretable and stable groups of NYC accidents using clustering in R while respecting the mixed-type nature of the data and the size of the dataset?

3 Why NYC Only

Restricting the work to NYC is methodologically useful for four reasons:

- the city-specific scope reduces plagiarism risk because the project becomes more focused and less generic
- the setting is geographically coherent
- borough-level interpretation is meaningful
- the dataset is still large enough to support serious analysis

The NYC subset contains **117,817** accident records. In this dataset, NYC is defined by the five borough counties:

Borough County	Rows
Bronx	27,308
Kings (Brooklyn)	21,234
New York (Manhattan)	21,989
Queens	41,480
Richmond (Staten Island)	5,806

4 Dataset Profile Relevant to Clustering

The NYC subset has enough variation to support clustering and enough volume to justify a substantial project.

4.1 Size and Severity Distribution

- Total rows: 117,817
- Severity 1: 65
- Severity 2: 85,847
- Severity 3: 29,550
- Severity 4: 2,355

This severity imbalance is important. It means severity should **not** be used as a clustering input variable if the goal is to interpret severity *after* the clusters are formed.

4.2 Useful Variables for Pattern Discovery

The dataset contains repeated roadway and environmental conditions that are appropriate for cluster discovery.

- `Crossing = TRUE` for about 21.83% of rows
- `Traffic_Signal = TRUE` for about 20.02% of rows

- `Junction = TRUE` for about 16.76% of rows
- `Stop = TRUE` for about 5.39% of rows
- `Railway = TRUE` for about 2.44% of rows
- highest accident hours include 06, 07, 08, 14, 15, 16, and 17
- common weather groups include Fair, Mostly Cloudy, Cloudy, Partly Cloudy, and Light Rain

4.3 Missingness

The most important missing-data observations are:

- `Weather_Condition`: about 1.03% missing
- `Temperature(F)`: about 0.69% missing
- `Humidity(%)`: about 0.69% missing
- `Pressure(in)`: about 1.65% missing
- `Visibility(mi)`: about 1.18% missing
- `Wind_Speed(mph)`: about 6.19% missing
- `Precipitation(in)`: about 23.87% missing

Implication:

- `Precipitation(in)` should not be a core clustering feature unless there is a clear imputation strategy
- moderate-missing numeric variables can be retained if the clustering method supports missing data appropriately

5 Research-Backed Method Selection

This is the core methodological section of the project.

5.1 Why Plain k-Means Should Not Be the Main Method

The standard R implementation of `kmeans()` expects numeric input. Our NYC accident data is not naturally numeric-only. It includes factors, binaries, time-derived categories, and mixed roadway conditions.

Therefore:

- plain k-means is not the best primary method for the full research workflow
- forcing all variables into crude numeric codes would damage interpretability
- k-means can still be used later as a reduced numeric-only comparison, but not as the main solution

5.2 Why Full Gower Plus Hierarchical Clustering or PAM Is Not the Main Method

The function `cluster::daisy()` is attractive because it supports mixed variable types through Gower-style dissimilarities. However, using full pairwise dissimilarities at NYC scale is not practical.

For 117,817 rows:

- pairwise distances = 6,940,363,836
- storing the distance matrix as doubles alone would require about 51.71 GiB

Therefore:

- full `daisy()` plus `agnes()` or `pam()` on all NYC observations is not recommended as the primary workflow
- this strategy may be used only on a sample for sensitivity analysis

5.3 Why CLARA Is Not the Best Primary Method

`cluster::clara()` is designed for large applications, which is an advantage. However, its documented input expects numeric or logical data. For this project, that means the mixed categorical structure would need extra encoding before clustering.

Therefore:

- CLARA is a reasonable fallback for a numeric or dummy-encoded workflow
- it is not the cleanest primary method for mixed accident records

5.4 Recommended Primary Method: k-Prototypes

The recommended primary method is `clustMixType::kproto()`.

Reasons:

- it is designed for mixed-type data
- it accepts numeric and factor variables together

- it supports `type = "gower"`, which range-normalizes numeric distances
- the package ecosystem supports validation and cluster stability analysis
- the associated R Journal paper explicitly argues that all-pairs mixed-distance approaches are not feasible for large datasets, whereas k-prototypes scales linearly in the number of observations

5.5 Recommended Secondary Method: DBSCAN

The recommended secondary method is `dbscan::dbscan()` for spatial hotspot discovery.

Reasons:

- accident hotspots often follow irregular road corridors and interchange shapes
- DBSCAN is designed for dense regions with arbitrary shape
- DBSCAN naturally labels sparse points as noise

5.6 Final Method Decision

The best overall project design is a two-layer clustering system:

1. **Scenario clustering**: use `kproto(type = "gower")` on mixed contextual variables
2. **Spatial clustering**: use `dbscan()` on projected coordinates

This is better than trying to force a single algorithm to answer every question.

6 Recommended Research Framing

The strongest topic statement is:

Cluster-Based Discovery of Accident Scenarios and Spatial Hotspots in New York City Using R

Recommended research questions:

- What latent accident scenario clusters exist in NYC when accidents are grouped by time, weather, and roadway context?
- How do these clusters differ across boroughs?
- Which discovered clusters contain a higher share of severe accidents?
- Where are the densest accident hotspot corridors in NYC?
- Do the hotspot areas correspond to particular scenario clusters?

7 What the Project Should and Should Not Do

7.1 What We Should Do

- use the NYC-only CSV as the official project dataset
- build one mixed-type scenario clustering pipeline in R
- build one separate spatial hotspot clustering pipeline in R
- exclude severity from the clustering input matrix and use it only afterward for interpretation
- engineer interpretable time features from `Start_Time`
- collapse rare weather categories into an `Other` group
- use borough mainly for reporting and comparison
- validate cluster quality and stability rather than choosing cluster counts arbitrarily

7.2 What We Should Avoid

- do not run plain k-means on the raw mixed dataset
- do not use `ID`, `Description`, `Street`, or raw text as the first clustering input
- do not include `Severity` as a clustering feature if the goal is to interpret severity later
- do not build a full all-pairs Gower distance matrix for all 117,817 rows
- do not overclaim causality
- do not rank boroughs as more dangerous without traffic exposure denominators

8 System Scope

8.1 In Scope

- ingestion of the NYC CSV
- cleaning and feature engineering in R
- mixed-type clustering for accident scenario discovery
- density-based spatial clustering for hotspot discovery
- cluster validation and stability assessment
- borough comparison of discovered clusters
- severity profiling after clustering
- exports, tables, and charts for the final report

8.2 Out of Scope

- causal inference
- real-time prediction
- live traffic feed integration
- deep learning on text descriptions
- national generalization

9 Stakeholders

The main stakeholders are:

- the student research team
- the course instructor
- transportation and safety audiences who may read the final report
- the class presentation audience

10 Functional Requirements

FR-1 Data ingestion. The system shall load the NYC accident CSV in R and preserve the original columns before preprocessing.

FR-2 Data cleaning. The system shall parse timestamps, handle missing values, and convert binary and categorical variables to suitable types.

FR-3 Time feature engineering. The system shall derive hour, hour band, weekday, weekend flag, month, season, and rush-hour status.

FR-4 Weather grouping. The system shall collapse rare weather categories into a smaller interpretable set.

FR-5 Scenario matrix creation. The system shall build a mixed-type feature matrix for scenario clustering.

FR-6 Leakage control. The system shall exclude `Severity`, IDs, free-text fields, and raw coordinates from the scenario clustering matrix.

FR-7 Scenario clustering. The system shall run `kproto()` with repeated starts to reduce unstable initializations.

FR-8 Cluster-count selection. The system shall evaluate candidate cluster counts, recommended as $k = 3$ through 8, using validation output.

FR-9 Stability analysis. The system shall perform bootstrap-based cluster stability checks.

FR-10 Cluster profiling. The system shall summarize each discovered cluster using dominant conditions, numeric summaries, borough composition, and severity composition.

FR-11 Spatial hotspot clustering. The system shall run DBSCAN on projected start coordinates and identify hotspot clusters and noise points.

FR-12 Export. The system shall export cluster-labeled outputs and summary tables for reporting.

11 Non-Functional Requirements

NFR-1 Reproducibility. The workflow shall be reproducible through R scripts or R Markdown with fixed random seeds where needed.

NFR-2 Interpretability. The clustering approach shall favor explainability over novelty.

NFR-3 Scalability. The main workflow shall operate on the full NYC subset without requiring a full pairwise distance matrix.

NFR-4 Maintainability. The code shall be split into readable preprocessing, clustering, validation, and export stages.

NFR-5 Academic defensibility. The algorithm choice shall be justified with source-backed reasons and clear methodological arguments.

NFR-6 Ethical reporting. Results shall be presented as exploratory pattern discovery, not proof of causation.

12 Data Requirements

12.1 Recommended Inputs for Scenario Clustering

Recommended features:

- `hour_band`
- `weekday`
- `weekend`
- `season`
- `rush_hour`
- `Sunrise_Sunset`
- `weather_group`
- `Temperature(F)`
- `Humidity(%)`

- `Pressure(in)`
- `Visibility(mi)`
- `Wind_Speed(mph)`
- `Crossing`
- `Junction`
- `Traffic_Signal`
- `Stop`
- `Railway`
- `Amenity`
- `Station`

12.2 Variables to Treat Carefully

- `Precipitation(in)`: high missingness, so exclude from the core model unless justified
- `Distance(mi)`: may reflect reporting behavior or event extent more than accident scenario type
- `County`: useful for post-cluster comparison, but not ideal as a dominant clustering input

12.3 Variables to Exclude From Scenario Clustering

- `Severity`
- `ID`
- `Description`
- `Street`
- `City`
- `Zipcode`
- `Start_Lat`
- `Start_Lng`
- `End_Lat`
- `End_Lng`

13 Detailed Analytical Design

13.1 Scenario Clustering

Goal:

- discover accident archetypes such as rush-hour intersection clusters, night visibility clusters, and junction-heavy corridor clusters

Recommended method settings:

- algorithm: `kproto()`
- distance type: `type = "gower"`
- candidate cluster counts: $k = 3 : 8$
- repeated starts: at least `nstart = 5`, preferably `10`
- missing-data handling: internal imputation or a clearly documented preprocessing step

13.2 Spatial Hotspot Clustering

Goal:

- discover dense accident corridors, interchange zones, and location-based hotspots

Recommended method settings:

- algorithm: `dbscan()`
- inputs: projected coordinates in approximate kilometer units
- parameter tuning: choose `eps` from a k-nearest-neighbor distance plot instead of guessing blindly
- inspect the percentage of noise points and the number of clusters

13.3 Interpretation Strategy

After clustering:

- join cluster labels back to the cleaned dataset
- compute severity distribution by cluster

- compute borough distribution by cluster
- compare day versus night composition by cluster
- identify dominant roadway-control patterns by cluster
- assign plain-English names to the discovered clusters

14 Validation Strategy

The project should not stop at obtaining cluster labels. It must show why the clusters are acceptable.

14.1 Validation for k-Prototypes

Use:

- `validation_kproto()` for internal validation across candidate values of k
- `stability_kproto()` for bootstrap stability checks
- `summary()` and `clprofiles()` for interpretation

Decision rule:

- choose a solution that balances interpretability, validation quality, and stability
- avoid cluster solutions dominated by tiny clusters unless they represent a meaningful rare-risk pattern

14.2 Validation for DBSCAN

Use:

- kNN distance plots for parameter selection
- hotspot size inspection
- map-based inspection of cluster shapes
- percentage of noise points

15 Use Cases

UC-1 Full workflow execution. A researcher runs the full preprocessing, scenario clustering, hotspot clustering, and export pipeline.

UC-2 Borough comparison. A researcher compares how boroughs are distributed across the discovered scenario clusters.

UC-3 Severity interpretation. A researcher evaluates which discovered clusters contain a larger proportion of severe accidents.

UC-4 Hotspot analysis. A researcher maps DBSCAN hotspot assignments and interprets dense accident zones.

16 Acceptance Criteria

The project is successful if it can:

- load and preprocess the NYC dataset reproducibly in R
- generate an interpretable mixed-type clustering solution
- justify the selected cluster count with validation output
- demonstrate cluster stability analysis
- produce a DBSCAN hotspot analysis
- connect cluster results back to severity and borough patterns
- generate report-ready tables and figures

17 Risks and Mitigations

Risk 1: Mixed-type distance is handled poorly.
Mitigation: use `kproto()` rather than forcing the data into plain k-means.

Risk 2: Location dominates the scenario clusters.
Mitigation: keep spatial clustering separate from scenario clustering.

Risk 3: Severity imbalance distorts the clusters.
Mitigation: do not cluster on severity directly.

Risk 4: Rare weather labels destabilize the solution.
Mitigation: collapse infrequent weather conditions into an `Other` category.

Risk 5: High precipitation missingness weakens the model.
Mitigation: exclude precipitation from the core design or justify imputation clearly.

Risk 6: The algorithm choice is challenged in review.
Mitigation: justify the method choice with official R documentation and package literature.

18 Phase-by-Phase Implementation Breakdown

This section translates the SRS into an execution plan that can be implemented step by step. Each phase should be completed before the next phase is finalized.

18.1 Phase 1: Repository Setup and Data Organization

Objective: prepare a clean project structure for Topic 3 and isolate the NYC clustering work from other project experiments.

Tasks:

- create a dedicated git branch for the third research topic
- keep the original large US dataset outside the tracked Topic 3 deliverables if it is not required for submission
- place the NYC subset, the SRS source, the compiled PDF, and the R clustering script in one topic-specific location
- remove temporary helper scripts and LaTeX build artifacts from the working directory

Deliverables:

- a clean Topic 3 folder or file set
- a dedicated git branch
- a clean git status before commit

Done criteria:

- only the files needed for Topic 3 remain as intended deliverables
- the branch is ready for commit and push

18.2 Phase 2: Data Preparation

Objective: build an analysis-ready NYC accident dataset in R.

Tasks:

- load the NYC CSV
- parse `Start_Time` and related time fields
- derive hour, hour band, weekday, weekend, month, season, and rush-hour indicators
- convert roadway flags and category variables into factors

- inspect missingness and decide which variables will be excluded, retained, or imputed
- collapse rare weather conditions into a smaller grouped factor

Deliverables:

- cleaned analysis-ready R data frame
- a documented feature list for clustering
- a short data-cleaning summary for the report

Done criteria:

- the dataset can be loaded and transformed reproducibly
- the final clustering input variables are fixed and justified

18.3 Phase 3: Scenario Clustering with k-Prototypes

Objective: discover mixed-type accident scenario groups across NYC.

Tasks:

- build the final mixed-type feature matrix
- exclude `Severity`, free-text columns, identifiers, and raw coordinates from the scenario matrix
- run candidate `kproto()` solutions for several values of k
- compare solutions using internal validation
- run repeated starts so the chosen clustering is not dependent on one weak initialization

Deliverables:

- candidate scenario clustering results
- validation output for the tested values of k
- a selected final scenario clustering model

Done criteria:

- one final scenario clustering solution is selected and justified
- the chosen clusters are interpretable and not trivially driven by one variable

18.4 Phase 4: Cluster Validation and Stability

Objective: prove that the selected scenario clusters are academically defensible.

Tasks:

- apply `validation_kproto()` to compare candidate solutions
- apply `stability_kproto()` using bootstrap-based resampling
- inspect cluster sizes to ensure the solution is not dominated by very small clusters
- profile the clusters to confirm that each group has a meaningful interpretation

Deliverables:

- validation results for the selected cluster count
- stability metrics
- cluster profile summaries

Done criteria:

- the final cluster choice is supported by validation and stability evidence
- each cluster can be described in plain language

18.5 Phase 5: Spatial Hotspot Clustering with DBSCAN

Objective: identify dense geographic accident hotspot zones in NYC.

Tasks:

- convert coordinates into projected distance units suitable for spatial clustering
- inspect k-nearest-neighbor distance behavior
- tune `eps` and `minPts`
- run DBSCAN on the projected coordinates
- separate hotspot clusters from noise points

Deliverables:

- final DBSCAN hotspot assignments
- hotspot counts and borough summaries
- parameter-selection justification

Done criteria:

- hotspot clusters are meaningful and not reduced to one giant cluster
- the level of noise is acceptable and explainable

18.6 Phase 6: Post-Clustering Interpretation

Objective: connect the discovered clusters back to real accident patterns.

Tasks:

- join scenario cluster labels back to the cleaned dataset
- compare severity distribution by scenario cluster
- compare borough composition by scenario cluster
- compare day versus night and weather composition by cluster
- compare hotspot clusters with scenario clusters where useful
- assign plain-English names to the major cluster types

Deliverables:

- cluster interpretation tables
- severity-by-cluster summaries
- borough-by-cluster summaries
- named cluster narratives for the report

Done criteria:

- the final report can explain what each cluster represents
- the interpretation is tied back to transportation and safety context

18.7 Phase 7: Reporting, Branching, and Submission

Objective: package the work cleanly for review and submission.

Tasks:

- finalize the SRS in LaTeX
- compile the final PDF

- keep the R script aligned with the documented workflow
- commit the Topic 3 files on a dedicated branch
- push the branch to the remote repository

Deliverables:

- final `.tex` SRS
- compiled PDF
- R clustering workflow
- pushed git branch for Topic 3

Done criteria:

- the branch contains only the intended Topic 3 deliverables
- the remote branch is available for sharing, review, or merge

19 Final Recommendation

If the project needs one strong and defensible direction, the best choice is:

Build an NYC accident clustering system in R with two complementary components:

- **k-prototypes** for mixed-type accident scenario discovery
- **DBSCAN** for spatial hotspot discovery

This direction is stronger than a simple descriptive analysis, more appropriate than plain k-means for the available variables, and more scalable than full-distance clustering on all NYC observations.

20 References

References

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